

Chapter 10: Applications of Trigonometry in Right-Angled Triangles

Name _____ () Class _____

ENDURING UNDERSTANDING

Students will be able to understand that:

1. Trigonometry enables us to model and solve problems involving indirect measurement.
2. Mathematical models simplify real-world situations while preserving essential relationships.
3. The way an object is viewed depends on the observer's position, but the underlying geometric relationships remain consistent.
4. Complex three-dimensional situations can be understood by analysing simpler two-dimensional representations.

ESSENTIAL QUESTIONS

1. How can trigonometry be used to determine measurements that are difficult or impossible to measure directly?
2. How can real-world situations be represented as right-angled triangles to solve problems?
3. How do angles of elevation and angles of depression help us describe and interpret the positions of objects?
4. How can complex three-dimensional situations be simplified into two-dimensional representations for problem solving?

LEARNING OBJECTIVES

At the end of the unit, students should be able to:

1. Find the unknown sides and angles in right-angled triangles.
2. Solve simple practical problems in two and three dimensions.

Reference Textbook

- Think! Mathematics New Syllabus Mathematics 8th Edition Textbook 2B. SL Education – Chapter 10

BIG IDEAS								
Student Learning Outcomes	Dimensions (Please tick the appropriate boxes)							
	FUNCTIONS	INVARIANCE	NOTATIONS	DIAGRAMS	MEASURES	EQUIVALENCE	PROPORTIONALITY	MODELS
	F	I	N	D	M	E	P	M
Pythagoras' Theorem			✓			✓	✓	
Trigonometric Ratios	✓				✓			

Level	Knowledge & Skills	I need help!	I can do on my own.	I can explain to others.
0	Identify the correct ratios of the lengths of a right-angled triangle for sine, cosine and tangent.			
1	Apply the Trigonometric ratios to find an unknown length and/or angle.			
1	Draw and annotate accurate diagrams from written descriptions and identify relevant right-angled triangles.			
1	Apply the Trigonometric ratios to solve simple geometrical problems involving angles of elevation and angles of depression.			
1	Represent three-dimensional situations using suitable two-dimensional views or cross-sections.			



It's a Bot time!

Need help along the way when it comes to Trigonometry?

Simply click the link or scan the QR code to chat with the Trigo Bot!

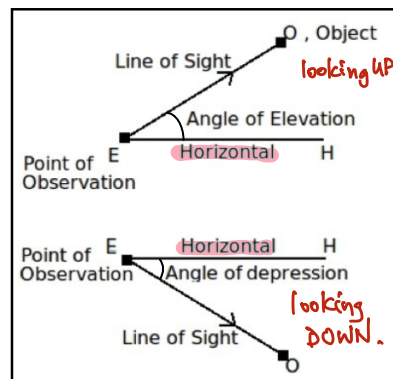
Visit: <https://student.magicschool.ai/s/join> and enter code: aroY6A



10.1 Angles of **Elevation** and **Depression**

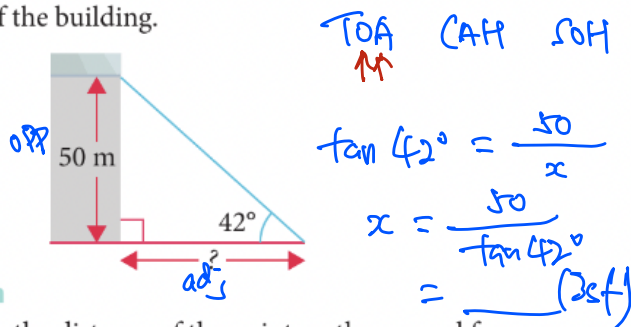
The line of sight is an imaginary line from the eye to a perceived object

- Suppose you are looking at an object in the distance.
- If the object is above you, then the angle of elevation is the angle your eyes look up.
- If the object is below you, the angle of depression is the angle your eyes look down.
- Angles of elevation and depression are measured from the horizontal.
- It is a common mistake not to measure the angle of depression from the horizontal.



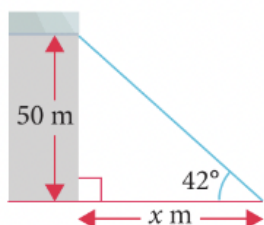
Worked Example 1: Angle of Elevation

A window of a building is 50 m above the ground. Given that the angle of elevation of the window from a point on the ground is 42° , find the distance of the point on the ground from the foot of the building.



Solution

Let x m be the distance of the point on the ground from the foot of the building.



$$\begin{aligned}\tan 42^\circ &= \frac{50}{x} \\ x \tan 42^\circ &= 50 \\ x &= \frac{50}{\tan 42^\circ} \\ &= 55.5 \text{ (to 3 s.f.)}\end{aligned}$$

$\therefore$ the distance of the point on the ground from the foot of the building is 55.5 m.

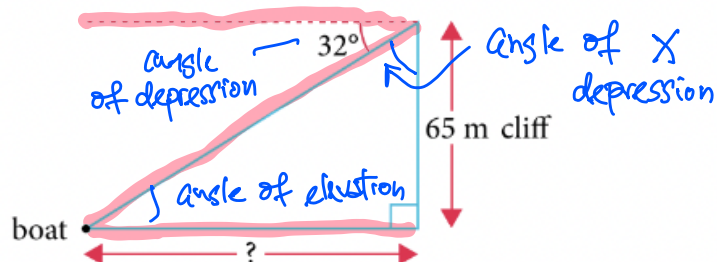
Big Idea

Diagrams

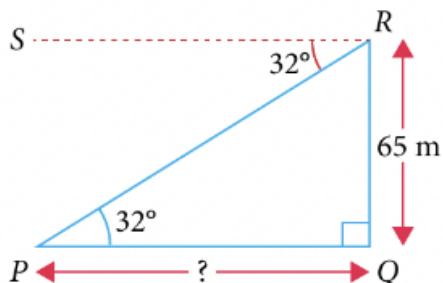
Diagrams help us visualise the relationships between the given mathematical objects. Here, with reference to the angle of 42° , the opposite side is 50 m. Assuming that the building is perpendicular to the ground, we can use the tangent ratio to find the length of the adjacent side.

Worked Example 2: Angle of Depression

A cliff is 65 m high. Given that the angle of depression from the top of the cliff to a boat is 32° , find the distance between the boat and the base of the cliff.



Solution



Problem-solving Tip

Label the vertices to represent the angles and lengths in the working.

Method 1:

$\angle RPQ = 32^\circ$ (alternate angles, SR is parallel to PQ)

$$\tan 32^\circ = \frac{65}{PQ}$$

$$PQ = \frac{65}{\tan 32^\circ}$$

$$= 104 \text{ m (to 3 s.f.)}$$

$\therefore$ the distance between the boat and the base of the cliff is 104 m.

Method 2:

$$\angle PRQ = 180^\circ - 90^\circ - 32^\circ \quad (\text{angle sum of a triangle})$$

$$= 58^\circ$$

$$\tan 58^\circ = \frac{PQ}{65}$$

$$PQ = 65 \tan 58^\circ$$

$$= 104 \text{ m (to 3 s.f.)}$$

$\therefore$ the distance between the boat and the base of the cliff is 104 m.

Reflection

Which method do you prefer? Why?

Example 1

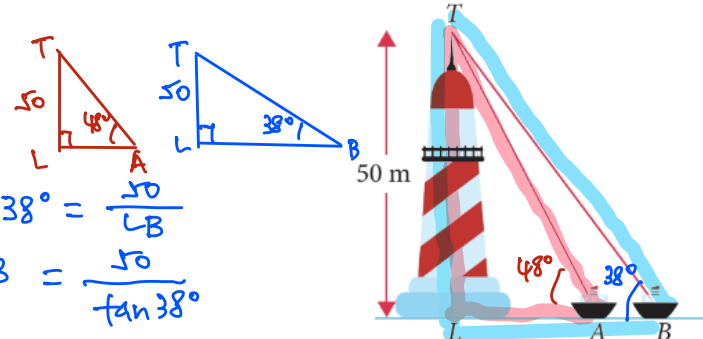
A lighthouse TL has a height of 50 m. The angles of elevation of the top of the lighthouse T from boat A and boat B are 48° and 38° respectively. Find the distance between boats A and B . — diff. AB .

$$AB = LB - LA$$

$$\tan 48^\circ = \frac{50}{LA} \quad \left| \quad \tan 38^\circ = \frac{50}{LB} \right.$$

$$LA = \frac{50}{\tan 48^\circ} \quad \left| \quad LB = \frac{50}{\tan 38^\circ} \right.$$

$$AB = \frac{50}{\tan 38^\circ} - \frac{50}{\tan 48^\circ}$$

$$= \underline{19.0 \text{ m}} \quad (3.s.f.)$$


Example 2

A boy whose eye level is 1.4 m above the ground finds that the angle of elevation of a bird on a tree is 40° . If the boy is 20 m from the tree,

- How high is the bird above the ground?
- If he moves a distance x m away from the tree, the angle of elevation becomes 35° . How far did he move from the original position?

(a) Let the height of the bird above ground be $h + 1.4$ m

$$\tan 40^\circ = \frac{h}{20}$$

$$h = 20 \tan 40^\circ$$

$$\therefore \text{height} = 20 \tan 40^\circ + 1.4 \text{ m}$$

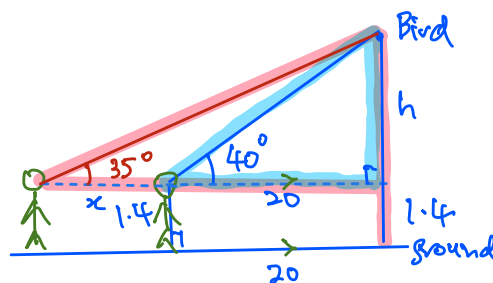
$$= \underline{18.2 \text{ m}} \quad (3.s.f.)$$

$$\tan 35^\circ = \frac{h}{x + 20}$$

$$x + 20 = \frac{h}{\tan 35^\circ}$$

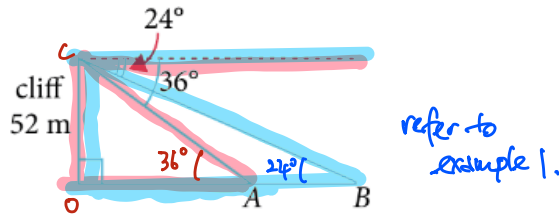
$$x = \frac{20 \tan 40^\circ}{\tan 35^\circ} - 20$$

$$= \underline{3.97 \text{ m}}$$



Example 3

The angles of depression from the top of a cliff, 52 m high, to two ships A and B due east of it are 36° and 24° respectively.



Calculate the distance between the two ships.

$$\tan 36^\circ = \frac{52}{OA} \quad \left| \quad \tan 24^\circ = \frac{52}{OB}\right.$$

$$OA = \frac{52}{\tan 36^\circ} \quad \left| \quad OB = \frac{52}{\tan 24^\circ}\right.$$

$$AB = OB - OA$$

$$= \frac{52}{\tan 24^\circ} - \frac{52}{\tan 36^\circ}$$

$$= \underline{45.2 \text{ m (3 s.f.)}}$$

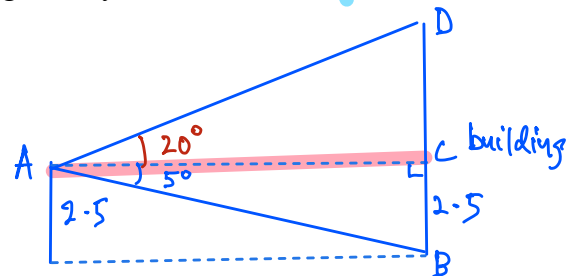
Example 4

An observer whose eye is 2.5 m above the ground observes the angle of depression of the foot of the building to be 5° , and the angle of elevation of the top to be 20° . Calculate the horizontal distance of the observer from the building, and the height of the building. Give your answers to the nearest metre.

$$\tan 5^\circ = \frac{2.5}{AC}$$

$$AC = \frac{2.5}{\tan 5^\circ}$$

$$= \underline{29 \text{ m (nearest m)}}$$



$$\text{Height} = BD = BC + CD$$

$$= 2.5 + \frac{2.5}{\tan 5^\circ} (\tan 20^\circ)$$

$$= \underline{13 \text{ m (nearest m)}}$$

$$\left. \begin{aligned} \tan 20^\circ &= \frac{CD}{AC} \\ CD &= AC \tan 20^\circ \\ &= \frac{2.5}{\tan 5^\circ} (\tan 20^\circ) \end{aligned} \right\}$$

10.2 Application of Trigonometric Ratios in 3-Dimension Problems

ICT Activity: Time to build! Download the Geogebra 3D app on your smartphone.

Recall your 3-D modelling skills applied during your Math Performance Task.

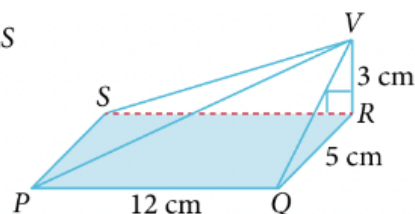
Now, build the models in the examples below, and bring them into your space with Augmented Reality!

Worked Example 3: 3-D problem involving pyramid

The figure shows a pyramid with a rectangular base $PQRS$ and vertex V vertically above R .

Given that $PQ = 12$ cm, $QR = 5$ cm and $VR = 3$ cm, find

- (i) $\angle VQR$, (ii) $\angle VPR$.

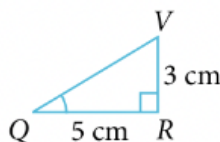


Solution

- (i) In $\triangle VQR$,

$$\tan \angle VQR = \frac{3}{5}$$

$$\begin{aligned}\angle VQR &= \tan^{-1} \frac{3}{5} \\ &= 31.0^\circ \text{ (to 1 d.p.)}\end{aligned}$$



Problem-solving Tip

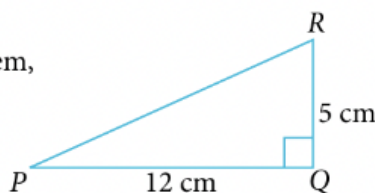
Reduce the three-dimensional to a problem in a plane by identifying the right-angled triangle that contains the side or the angle to be found.

- (ii) In $\triangle PQR$, $\angle PQR = 90^\circ$.

Using Pythagoras' Theorem,

$$\begin{aligned}PR^2 &= PQ^2 + QR^2 \\ &= 12^2 + 5^2 \\ &= 144 + 25 \\ &= 169\end{aligned}$$

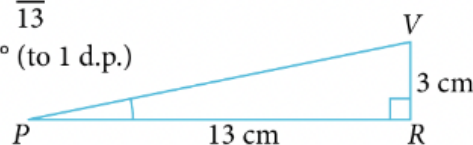
$$\begin{aligned}PR &= \sqrt{169} \\ &= 13 \text{ cm}\end{aligned}$$



In $\triangle VRP$, $\angle VRP = 90^\circ$. VR is the normal to the plane $PQRS$

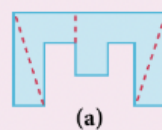
$$\tan \angle VPR = \frac{3}{13}$$

$$\begin{aligned}\angle VPR &= \tan^{-1} \frac{3}{13} \\ &= 13.0^\circ \text{ (to 1 d.p.)}\end{aligned}$$

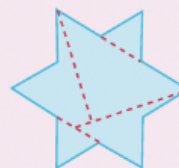


Just For Fun

Two shapes are as shown.



(a)



(b)

Cut each shape along the dotted lines and use the pieces to form two squares.

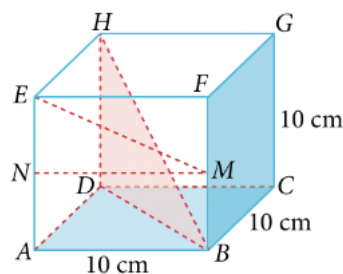
Worked Example 4: 3-D problem involving cube

The figure shows a cube of length 10 cm.

M and N are the midpoints of BF and AE respectively.

Find

- (i) $\angle MEN$,
 (ii) $\angle EMN$,
 (iii) $\angle MEF$,
 (iv) $\angle HBD$.



■ Solution

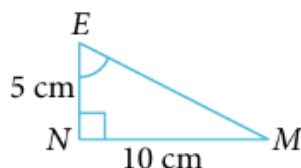
- (i) In $\triangle MEN$,

$$\tan \angle MEN = \frac{10}{5}$$

$$= 2$$

$$\angle MEN = \tan^{-1} 2$$

$$= 63.4^\circ \text{ (to 1 d.p.)}$$



- (ii) **Method 1:**

In $\triangle MEN$,

$$\tan \angle EMN = \frac{5}{10}$$

$$= \frac{1}{2}$$

$$\angle EMN = \tan^{-1} \frac{1}{2}$$

$$= 26.6^\circ \text{ (to 1 d.p.)}$$

Method 2:

$$\angle EMN = 180^\circ - \angle ENM - \angle MEN \text{ (angle sum of a triangle)}$$

$$= 180^\circ - 90^\circ - 63.435^\circ$$

$$= 26.6^\circ \text{ (to 1 d.p.)}$$

- (iii) $\angle MEF = \angle EMN$ (alternate angles, EF is parallel to NM)
 $\therefore \angle MEF = 26.6^\circ$

- (iv) In $\triangle BCD$, $\angle BCD = 90^\circ$.

Using Pythagoras' Theorem,

$$DB^2 = BC^2 + DC^2$$

$$= 10^2 + 10^2$$

$$= 100 + 100$$

$$= 200$$

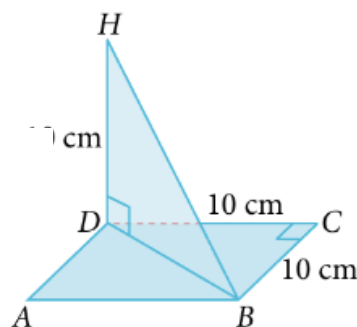
$$DB = \sqrt{200}$$

In $\triangle HBD$, $\angle HDB = 90^\circ$.

$$\tan \angle HBD = \frac{10}{\sqrt{200}}$$

$$\angle HBD = \tan^{-1} \frac{10}{\sqrt{200}}$$

$$= 35.3^\circ \text{ (to 1 d.p.)}$$

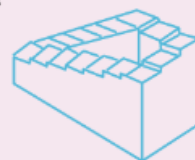


Reflection

- (ii) Which method do you prefer? Why?

Just For Fun

Is it possible to construct the following three-dimensional objects from the solids that you know?



(a)



(b)



(c)

Example 5

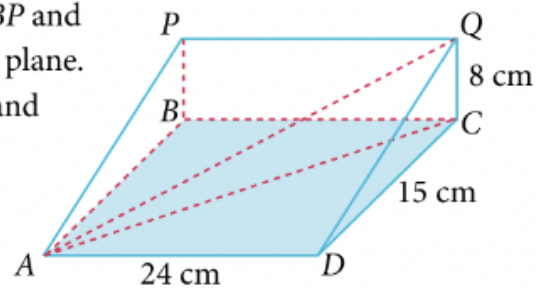
The figure shows a wedge with a horizontal base $ABCD$ and a vertical face $PQCB$.

$APQD$ is a rectangular sloping surface and $\triangle ABP$ and $\triangle DCQ$ are right-angled triangles in the vertical plane.

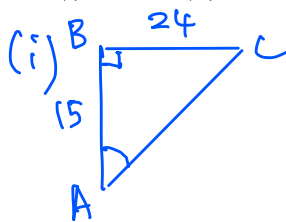
Given that $CQ = BP = 8$ cm, $DC = AB = 15$ cm and

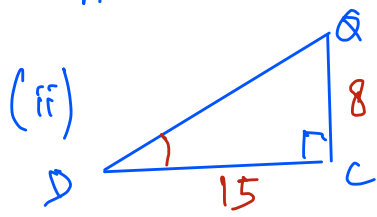
$AD = BC = PQ = 24$ cm, find

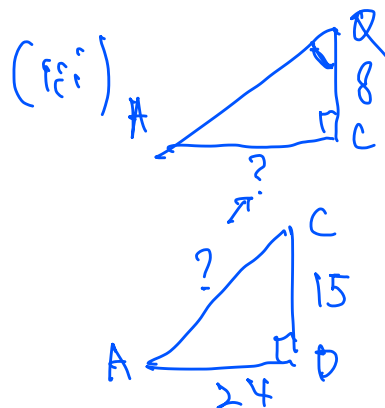
- (i) $\angle BAC$, — look at $\triangle BAC$.
- (ii) $\angle CDQ$, — look at $\triangle CDQ$
- (iii) $\angle AQC$. — look at $\triangle AQC$



[Ans: (i) 58.0° , (ii) 28.1° , 74.2°]

(i)  $\tan BAC = \frac{24}{15}$ use $\boxed{\tan^{-1}}$ calculator
 $\angle BAC = \underline{\hspace{2cm}}$
 $= \underline{58.0^\circ} \text{ (1 d.p.)}$

(ii)  $\tan CDQ = \frac{8}{15}$
 $\angle CDQ = \underline{\hspace{2cm}}$
 $= \underline{28.1^\circ} \text{ (1 d.p.)}$



$$AC = \sqrt{24^2 + 15^2}$$

$$= \sqrt{801}$$

$$\tan AQC = \frac{AC}{8}$$

$$= \frac{\sqrt{801}}{8}$$

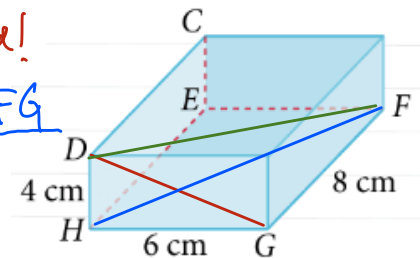
$$\angle AQC = \underline{74.2^\circ} \text{ (1 d.p.)}$$

Example 6

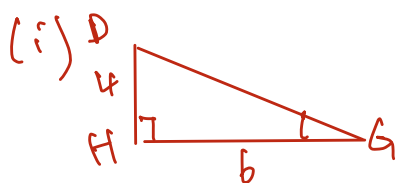
A rectangular block of sugar has a horizontal base $EFGH$. The corners C and D are vertically above E and H respectively.

$DH = 4$ cm, $GH = 6$ cm and $FG = 8$ cm. Find

- (i) $\angle DGH$, $\rightarrow$ Is $\triangle DGH$ right-angled? Yes, good!
 (ii) HF , $\rightarrow$ look for right-angled $\triangle$ involving $HF \rightarrow \triangle HFG$
 (iii) $\angle DFH$, $\rightarrow$ Is $\triangle DFH$ right-angled?



[Ans: (i) 33.7° , (ii) 10 cm, (iii) 21.8°]

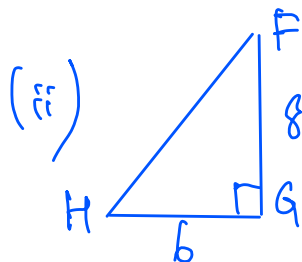


$$\tan DGH = \frac{4}{6}$$

$$\angle DGH = \tan^{-1}\left(\frac{4}{6}\right)$$

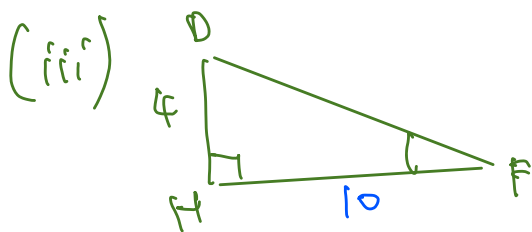
calculator use $\boxed{\tan^{-1}}$ $\tan^{-1}\left(\frac{4}{6}\right)$

$$= 33.7^\circ (1 \text{ d.p.})$$



$$HF = \sqrt{6^2 + 8^2}$$

$$= 10 \text{ cm.}$$



$$\tan DFH = \frac{4}{10}$$

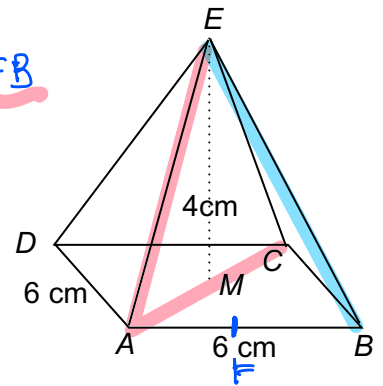
$$\angle DFH = \tan^{-1}\left(\frac{4}{10}\right)$$

$$= 21.8^\circ (1 \text{ d.p.})$$

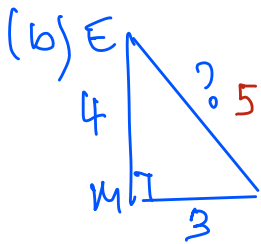
Example 7

The diagram shows a square pyramid $ABCDE$ with E 4 cm vertically above M , the mid-point of the base, and $ABCD$ is of side 6 cm.

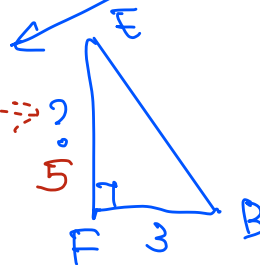
- (a) obvious State the vertical height of the pyramid.
 (b) Find the length of EB . look at $\triangle EMB$ or $\triangle EFB$
 (c) Find the angle AEM . $\triangle AEM$
 (d) Is $\angle AEM$ the same as $\angle BEM$? Justify your answer.
 (e) Find the angle between AE and AC .



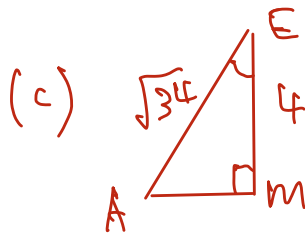
(a) 4 cm. look for right angled Δ .



$$EF = \sqrt{3^2 + 4^2} \\ = 5 \text{ cm}$$



$$\therefore EB = \sqrt{5^2 + 3^2} \\ = \sqrt{34} \\ = 5.83 \text{ cm (3 s.f.)}$$



$$\cos AEM = \frac{4}{\sqrt{34}} \\ \angle AEM = 46.686 \\ = \underline{46.7^\circ} \text{ (1 d.p.)}$$

(d) $\triangle AEM$ and $\triangle BEM$

EM is common

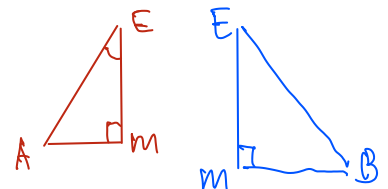
$$AM = BM$$

$$\angle AME = \angle BME = 90^\circ$$

$$\triangle AEM \equiv \triangle BEM \text{ (SAS Congruence Test)}$$

$$\therefore \angle AEM = \angle BEM$$

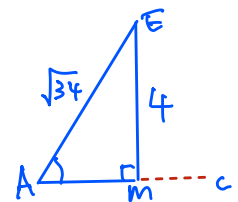
SSS, AAS, SAS, RHS



(e) $\angle EAC = \angle EAM$

$$\sin EAM = \frac{4}{\sqrt{34}}$$

$$\therefore \angle EAC = \angle EAM \\ = \underline{43.3^\circ} \text{ (1 d.p.)}$$



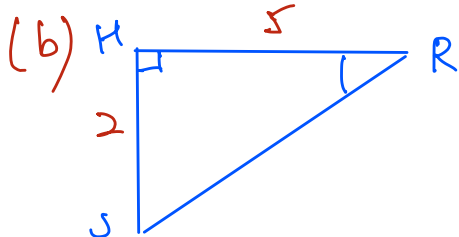
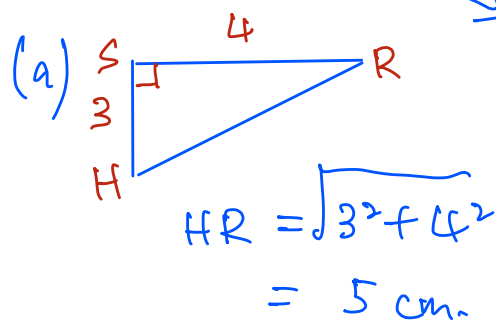
Example 8

H and J are respectively the midpoints of the sides PS and AD of the cuboid with dimensions 4 cm, 6 cm and 2 cm.

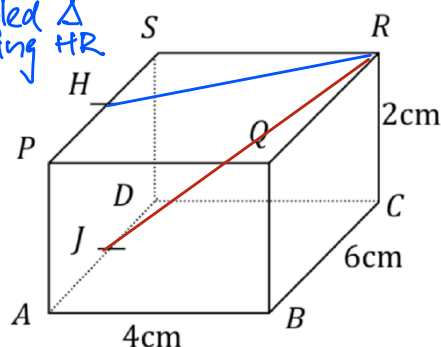
(a) Find the length of HR .

(b) Find the length of JR .

(c) Calculate angle JRH .



(c) $\tan JRH = \frac{2}{5}$
 $\angle JRH = 21.8^\circ$ (1 d.p.)

**ASSIGNMENTS****Assignment 10.1**

Think! Mathematics New Syllabus Mathematics 8th Edition Textbook 2B pg 69 -- 70

Exercise 10B

Tier A: Q4

Tier B: Q5, 6, 7

Assignment 10.2

Think! Mathematics New Syllabus Mathematics 8th Edition Textbook 2B pg 73 -- 74

Tier A: Q2

Tier B: Q3, 4, 5

Tier C: Q6, 8

Assignment 10.3

Think! Mathematics New Syllabus Mathematics 8th Edition Textbook 2B pg 80 -- 81

Tier A: Q5, 6

Tier B: Q7, 12

Tier C: Q13

$$\boxed{\tan} A = \left(\frac{3}{8} \right)$$

$$\angle A = \tan^{-1} \left(\frac{3}{8} \right)$$

$$x^2 = \frac{1}{8}$$

$$x = \frac{\sqrt{1}}{8} \times \sqrt{\frac{1}{8}}$$